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node.js



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INTERVIEW QUESTIONS

The Key Concepts To Master



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1 What is Node.js?**Answer:**

Node.js is a JavaScript runtime engine used for executing JavaScript code outside the browser, commonly used to build scalable backend applications.

2 What is the difference between Node.js and JavaScript?**Answer:**

JavaScript is a scripting language, while Node.js is a runtime environment that allows JavaScript to run on the server side.

3 Is Node.js single-threaded?**Answer:**

Yes, Node.js is single-threaded but uses event-driven architecture and non-blocking I/O to handle multiple requests efficiently.

4 What kind of API functions does Node.js support?**Answer:**

Node.js supports both synchronous (blocking) and asynchronous (non-blocking) API functions.

5 What is a module in Node.js?**Answer:**

A module in Node.js is a block of code that provides specific functionality, which can be reused across different parts of an application.



6

What is npm and its advantages?**Answer:**

npm is the default package manager for Node.js, offering benefits like dependency management, version control, and a centralized repository.

7

What is middleware?**Answer:**

Middleware functions execute between the request and response cycle, performing tasks like logging, authentication, and data processing.

8

How does Node.js handle concurrency despite being single-threaded?**Answer:**

Node.js handles concurrency through asynchronous, non-blocking operations, allowing multiple tasks to run simultaneously within a single thread.

9

What is control flow in Node.js?**Answer:**

Control flow refers to the order in which code statements and functions are executed, managing asynchronous operations and error handling.

10

What is the event loop in Node.js?**Answer:**

The event loop is a mechanism that processes asynchronous tasks in a single thread by continuously checking for and executing callback functions.



11

What are the main disadvantages of Node.js?**Answer:**

Disadvantages include its single-threaded nature, preference for NoSQL databases, and rapid API changes that can cause instability.

12

What is REPL in Node.js?**Answer:**

REPL stands for Read, Evaluate, Print, and Loop; it's an interactive environment for executing Node.js code and debugging.

13

How do you import a module in Node.js?**Answer:**

Use the `require()` function to import external modules, storing the result in a variable for use in the application.

14

What is the difference between Node.js and AJAX?**Answer:**

Node.js is a server-side runtime environment, while AJAX is a client-side technique for asynchronously updating parts of a web page.

15

What is package.json in Node.js?**Answer:**

package.json is a metadata file in Node.js that contains information about the project, such as dependencies, scripts, and version.



16

What is the most popular Node.js framework?**Answer:**

The most popular Node.js framework is Express.js, known for its scalability and minimalistic approach to building web applications.

17

What are promises in Node.js?**Answer:**

Promises in Node.js are objects that handle asynchronous operations, providing a cleaner alternative to callback functions.

18

What is event-driven programming in Node.js?**Answer:**

Event-driven programming synchronizes multiple events using event loops and callback functions to simplify program flow.

19

What is a buffer in Node.js?**Answer:**

A buffer is a temporary storage space for binary data, allowing Node.js to handle raw data directly.

20

What are streams in Node.js?**Answer:**

Streams are objects used to handle continuous data flows, allowing for efficient reading and writing of data.



21

Explain the crypto module in Node.js.**Answer:**

The crypto module provides cryptographic functionality, such as encryption, decryption, and hashing of data.

22

What is callback hell?**Answer:**

Callback hell refers to the problematic situation caused by deeply nested callbacks, making code difficult to read and maintain.

23

Explain the use of the timers module in Node.js.**Answer:**

The timers module allows execution of code after a specified delay or immediately in the next event loop cycle using `setTimeout()` and `setImmediate()`.

24

What is the difference between `setImmediate()` and `process.nextTick()`?**Answer:**

`process.nextTick()` executes callbacks at the start of the next event loop, while `setImmediate()` executes them at the end of the current event loop.

25

What is the difference between `setTimeout()` and `setImmediate()`?**Answer:**

`setTimeout()` schedules a callback after a specified delay, whereas `setImmediate()` executes it immediately after I/O events.



26

What is the difference between `spawn()` and `fork()`?**Answer:**

`spawn()` runs a new process from the command line, while `fork()` creates a new instance of the existing process to perform parallel tasks.

27

Explain the use of the passport module in Node.js.**Answer:**

The passport module adds authentication features to applications, supporting various sign-in methods including OAuth and local strategies.

28

What is fork in Node.js?**Answer:**

Fork is a method to create child processes that allow parallel execution of tasks in Node.js.

29

What are three methods to avoid callback hell?**Answer:**

To avoid callback hell, use `async/await`, `promises`, or `generators` — each provides a cleaner way to handle async operations.

30

What is body-parser in Node.js?**Answer:**

Body-parser is middleware that parses incoming request bodies before handling them in Node.js applications.



31

What is CORS in Node.js?**Answer:**

CORS stands for Cross-Origin Resource Sharing, allowing restricted resources on a web page to be requested from another domain.

32

Explain the tls module in Node.js.**Answer:**

The tls module provides an implementation of TLS and SSL protocols to establish secure network connections.

33

What is a cluster in Node.js?**Answer:**

A cluster allows Node.js to utilize multiple CPU cores by creating child processes that share the same server port.

34

How do you manage sessions in Node.js?**Answer:**

Sessions in Node.js can be managed using the express-session module, which stores session data on the server.

35

Explain the types of streams in Node.js.**Answer:**

Types of streams include readable, writable, duplex (both read and write), and transform (modifies data while streaming).



36 How can we implement authentication and authorization in Node.js?**Answer:**

Use packages like Passport for authentication and JWT for managing tokens to implement security in Node.js applications.

37 What packages are used for file uploading in Node.js?**Answer:**

Multer is a popular middleware used for handling file uploads in Node.js applications.

38 How do you handle database connections in Node.js?**Answer:**

Database connections in Node.js are managed using drivers like MySQL and libraries like Mongoose for MongoDB.

39 How do you read command line arguments in Node.js?**Answer:**

Use the `process.argv` array to access command-line arguments passed when running a Node.js application.

40 What are child processes in Node.js?**Answer:**

Child processes allow Node.js to handle multiple tasks concurrently by creating subprocesses that can run independently.



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